

PAPER • OPEN ACCESS

Temporal Variability and Pollution Loading of Wet Market Wastewater Discharges in Urban Catchments

To cite this article: Idzumi Mohamed Hasnan *et al* 2026 *IOP Conf. Ser.: Earth Environ. Sci.* **1665** 012012

You may also like

- [The Detection of Microbial Contamination on Goat Meat from Traditional Market in Bandar Lampung City](#)
N Riffiandi, G G Maradon and V R Pertiwi
- [Study on the Evolution Stages and Mechanism of China's Unified National Electricity Market](#)
Yan Yang, Shuo Liu, Mengxin Lan et al.
- [Relating the model to the market](#)
Nick Webber

View the [article online](#) for updates and enhancements.

Temporal Variability and Pollution Loading of Wet Market Wastewater Discharges in Urban Catchments

Author: Idzumi Mohamed Hasnan^{1a*}, Nur Muna Mazlan¹, Aleesya Ellanie Zambre¹

¹Aurecon Lestari Sdn. Bhd., Kuala Lumpur, Malaysia

*E-mail: idzumi.mohamedhasnan@aurecongroup.com

Abstract. Wet markets are recognised as important non-domestic sources of organic rich sullage. However, information on their discharge characteristics and temporal variability remains limited. This study presents a discharge profiling and pollution loading assessment of Bangi Wet Market and Mantin Wet Market, located within the Sg. Semenyih and Sg. Beranang catchments, respectively. A structured sampling programme was conducted over a full operational cycle to capture temporal variations associated with market activities. Key physicochemical and microbiological parameters analysed include pH, temperature, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), suspended solids (SS), ammoniacal nitrogen (NH₃-N), oil and grease, and faecal coliform. Results indicate that wet market discharges consistently exhibited high concentrations of BOD, COD, NH₃-N, and SS, with pronounced temporal fluctuations corresponding to peak operational periods. Mantin Wet Market, characterised by intensive chicken slaughtering activities, recorded substantially higher pollutant loadings with elevated concentrations of BOD, COD, NH₃-N and faecal coliform counts. Potential impacts on downstream water bodies were assessed, particularly during peak discharge. The findings highlight the lack of effective treatment and regulatory controls for wet market sullage and underscore the need for targeted management measures, improved waste handling practices, and appropriate discharge standards to mitigate impacts on receiving waters.

Keywords: wet market, discharge profiling, water quality impact, pollutant loading.

1. Introduction

The discharge of wet market wastewater containing organic and inorganic pollutants into the environment contributes significantly to water pollution [1]. The release of this wastewater, into aquatic ecosystems is a major environmental concern due to its potential threat to ecological and public health. Solid wastes form another component of waste generated from wet markets [2-4] and are predominantly being treated by landfilling [5].

In this study, key physicochemical and microbiological parameters including pH, temperature, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD),

^a To whom any correspondence should be addressed.



suspended solids (SS), ammoniacal nitrogen ($\text{NH}_3\text{-N}$), oil and grease, and faecal coliform are investigated to capture temporal variations associated with market activities. A discharge profiling and pollution loading assessment of two wet markets located in urban settings is performed. Potential impacts on receiving water bodies are assessed, including pollutant loading and dilution effects, while highlighting the need for effective treatment, waste handling practices and regulatory controls.

2. Literature Review

2.1 Compliance Requirements for Wet Market Wastewater Discharge in Malaysia

To date, there are no gazetted standards for wastewater discharge from either the Local Government Act, 1976 or the Environment Quality Act 1974 for wet markets to govern discharge limits. However, local authority bylaws and regulations are an interim measure for enforcing control measures on wet market wastewater discharge. For example, the discharge of wastewater from wet markets in Negeri Sembilan governed through local bylaws under the Local Government Act 1976 which regulates cleanliness, drainage, waste disposal, and business licensing in markets. Wastewater, poultry, fish and food residues cannot be discharged into public drains or walkways. Wastewater shall only be discharged into designated drainage systems or floor traps provided by the council.

For purposes of this study, the Environmental Quality (Industrial Effluent) Regulations 2009 (IER) will be used as a baseline to measure compliance of the wet market wastewater discharge. The IER is selected due to its intended purpose for regulating industrial effluent discharge to receiving water bodies and considering wet markets are commercial or trade premises. Hence, the wet market wastewater is hereby classified as industrial (trade) effluent, despite its biodegradable properties.

2.2 Typical Characteristics of Wet Market Wastewater Discharge

Wet markets generate sullage, primarily from poultry slaughtering activities, fish cleaning processes, and routine washing of market floors and stalls. Findings from Zulkifli et al. [1] indicates that approximately 50% of the 865 wet markets nationwide have ceased on-site slaughtering activities. Nevertheless, slaughtering operations are still conducted at about 21% of wet markets, while the activity status of the remaining 29% could not be ascertained.

Table 1 illustrates typical characteristics of wet market discharge parameters for four wet markets in Seremban [1]. The comparison of wet market sullage characteristics with the IER, reveals widespread non-compliance with both Standard A and Standard B discharge limits.

The results demonstrate that all sampled wet markets fail to comply with IER Standard A discharge requirements, and the majority exceed IER Standard B limits for key parameters including BOD, TSS, oil and grease, and ammoniacal nitrogen. These exceedances highlight systemic deficiencies in wastewater and solid waste management at wet markets, particularly in facilities that continue on-site slaughtering and rely on direct discharge to drains without adequate treatment.

Table 1. Characteristics of Wet Market Discharge Parameters and National Standards for Industrial Effluent Discharge.

Parameter	Unit	Pasar Seremban	Pasar Ampangan	Pasar Rasah Jaya	Pasar Seremban Jaya	Std. A	Std. B
pH	mg/L	4.7–6.3	5.6–5.8	5.8	4.4	6.0–9.0	5.5–9.0
COD	mg/L	656–1440	381–560	763	397	–	–
BOD	mg/L	137–295	71–122	153	68	20	50
TSS	mg/L	192–544	60–122	900	250	50	100
O&G	mg/L	40–72	13–43	10	28	1	10
NH₃-N	mg/L	26.4–57.7	2.6–14.6	43.8	7.3	10	20
<i>E. coli</i>	CFU/100 mL	280–1600	50–170	900	220	–	–
Nitrite NO₂	mg/L	0.115–0.167	0.094–0.291	0.152	0.112	–	–
Nitrate NO₃	mg/L	2.44–6.49	1.98–2.24	4.92	6.53	–	–
Phosphate PO₄	mg/L	2.2–39.6	ND–22.2	44.5	27.7	–	–
Total Nitrogen	mg/L	55.2–72.9	30.3–37.3	61.1	41.6	–	–

Note: Std. A and Std. B refer to Standard A and Standard B, respectively, under the Environmental Quality (Industrial Effluent) Regulations 2009

2.3 Waste Management Technologies for Treating Wet Market Discharge

According to statistics reported by the Ministry of Housing and Local Government (MHLG), only 11% of wet markets in Malaysia are equipped with wastewater treatment systems, whereas 61% operate without any form of treatment. The prevailing practice involves the direct discharge of untreated wastewater into drainage networks or receiving water bodies, with reliance on natural dilution facilitated by historically high rainfall in the region. Consequently, there is an important need to adopt alternative, sustainable, and effective treatment strategies. Several treatment, recycling and resource recovery solutions for wet market biowaste management are presented next.

Direct use of wet market biowastes including vegetable trimmings, fish residues, and selected animal byproducts, has been practiced in various regions through land application as soil amendments, animal feeding, and vermiculture. International guidance and reviews emphasise that direct use is only acceptable where feedstocks are source segregated, safely handled, and

subjected to minimum hygienisation measures such as thermal treatment, controlled fermentation, or biological stabilisation prior to reuse [6,7].

Biological treatment of wet market biowastes relies on microbial or biotic processes to convert putrescible residues from meat, fish and produce stalls into safer, value added products.. Common options include composting, black soldier fly larvae bioconversion, anaerobic digestion and microalgal cultivation. Composting and co-composting can stabilise slaughtering wastes such as manure, litter and feathers when moisture, aeration and carbon nitrogen balance are well controlled, while also reducing pathogen risks and producing compost suitable for soil application [8,9]. Black soldier fly larvae treatment provides rapid mass reduction of protein rich wastes and produces larval biomass and nutrient rich frass [10,11]. Anaerobic digestion is particularly suitable for high strength wastes such as blood and offal, enabling COD reduction and biogas recovery, especially when co-digested with fruit and vegetable wastes to improve process stability [12-15]. Microalgal cultivation on pre-treated wastewater can further remove nutrients while producing biomass for potential feed or fertiliser applications [16].

Physico-chemical treatment is primarily used to separate and condition lipid and solids rich wastes before downstream biological treatment. Grease interceptors are essential for capturing fats, oils and grease, while coagulation–flocculation and dissolved air flotation can reduce suspended solids and COD loadings [17,18]. Chemical precipitation, electrocoagulation and chemical conditioning may also be applied for nutrient removal, stream stabilisation and resource recovery [19-21]. Although electrocoagulation has demonstrated high pollutant removal efficiencies, its implementation in small wet markets may be constrained by electricity consumption, electrode replacement requirements, and operational costs. Consequently, simpler source-control measures such as grease traps and waste segregation are likely to be more practical and cost-effective for smaller facilities, while electrocoagulation may be more suitable for larger centralized markets with higher wastewater volumes [18,19].

Thermochemical processes such as pyrolysis, gasification and incineration provide options for sanitising and valorising pathogen bearing, high strength residues while reducing landfill dependence [7,22,23]. Nevertheless, source control remains fundamental; dry cleaning before washdown, segregation of blood and offal, sealed storage of organic wastes, and improved sanitation and drainage infrastructure are important measures to reduce pollutant loads and improve the effectiveness of downstream treatment systems [6,24].

3. Methodology

3.1 Wet Market Selection

Two wet markets were sampled for this study. The first, Bangi Wet Market is located in Pekan Bangi Lama within the Sg. Semenyih catchment area. The sampling location for Bangi wet market is shown in Figure 1.

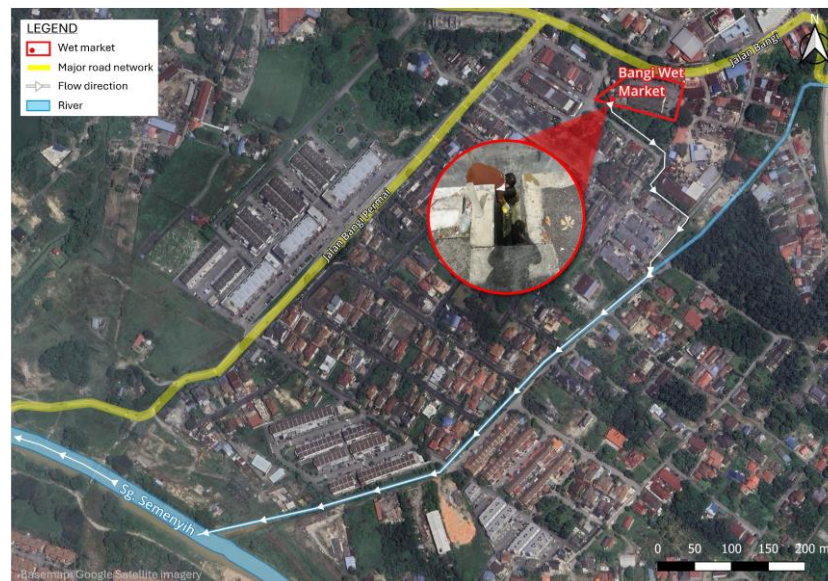


Figure 1. Location of Bangi Wet Market, its Wastewater Discharge to Sg. Semenyih and Sampling Location

The market comprises four operating stalls, each associated with distinct commercial activities that influence the quantity and characteristics of the wastewater generated. These stalls include the sale of coconut milk, poultry, fish and seafood, dried pantry staples (such as pulses, nuts, dried fruits, spices, and oils), and seasonal locally sourced vegetables. Among these, stalls involving poultry, fish, and seafood handling are expected to contribute higher organic loads, suspended solids, and faecal contamination due to frequent washing and handling of raw animal products. In contrast, stalls selling dried goods are anticipated to generate comparatively lower wastewater volumes and pollutant loads.



Figure 2. Location of Mantin Wet Market, its Wastewater Discharge to Sg. Setul (a tributary of Sg. Beranang) and Sampling Location

The second, Mantin Wet Market is located along Lorong Sri Keledang within the Sg. Beranang catchment area. The sampling location of Mantin wet market is shown in Figure 2. The market comprises nine operating stalls selling vegetables and poultry. The dominant activity at Mantin Wet Market is chicken slaughtering which generate wastewater with elevated organic matter, suspended solids, fats, and microbial content.

Poultry slaughtering operations typically commence as early as 6:00 a.m. and continue until approximately 10:00 a.m. In contrast, wastewater generated from vegetable stalls is mainly associated with produce washing and general cleaning and is expected to contribute comparatively lower organic and microbial loads.

3.2 Wet Market Discharge Profiling

The primary objective of the wet market discharge profiling is to characterise temporal trends in wastewater discharge quality and pollutant loading throughout the daily operating cycle. The discharge profiling study was conducted for Bangi and Mantin Wet Markets over a two day period. Wastewater samples were collected at 30 minute intervals to capture temporal variations in discharge characteristics during market operation hours. The parameters analysed included pH, temperature, dissolved oxygen (DO), biochemical oxygen demand (BOD₅), chemical oxygen demand (COD), ammoniacal nitrogen (NH₃-N), suspended solids (SS), oil and grease, and faecal coliform.

3.3 Mixing Model for Assessing Impacts on Receiving Waters

A steady state complete mixing model was applied to evaluate the potential impact of wet market wastewater discharge on the receiving waterbody (river) under normal and worst case hydrological conditions.

The expected final concentrations of each parameter, following mixing of the discharged wastewater with the receiving water is calculated as:

$$C_m = \frac{Q_b C_b + Q_d C_d}{Q_m} \quad (1)$$

Where:

C_b =	parameter concentration of baseline	Q_b =	flow rate of baseline
C_d =	parameter concentration of wastewater discharge	Q_d =	flow rate of wastewater discharge
C_m =	predicted parameter concentration after mixing	Q_m =	flow rate of downstream after mixing

Two modelling scenarios were defined and evaluated:

- Normal condition, representing wet market discharge during typical river flow conditions observed during the field monitoring period; and
- Worst case condition, representing discharge during critical low flow conditions with reduced dilution capacity.

The input conditions adopted for each scenario are summarised in Table 2.

Table 2. Input Parameters Adopted for Water Quality Mixing Model

Item	Scenario 1: Normal condition (typical flow)	Scenario 2: Worst case (low flow)
Q_b (m ³ /s)	8.09	0.183 (7Q10 Hydrological Procedure No. 12 by DID)*
C_b (mg/L)	Measured concentrations from first sampling round	Maximum from two sampling rounds for SS, COD, BOD ₅ , NH ₃ -N; minimum for DO
Q_d (m ³ /s)	Measured discharge from Mantin Wet Market point source sampling	Measured discharge from Mantin Wet Market point source sampling
C_d (mg/L)	Pollutant concentrations from wet market profiling at peak slaughtering period	Pollutant concentrations from wet market profiling at peak slaughtering period
Parameters assessed	DO, BOD ₅ , COD, SS, NH ₃ -N	DO, BOD ₅ , COD, SS, NH ₃ -N

Note: *7Q10 values were estimated based on Hydrological Procedure No. 12: Magnitude and Frequency of Low Flows in Peninsular Malaysia published by the Department of Irrigation and Drainage (DID) Malaysia.

For the scenarios given, the following assumptions and criteria were adopted:

- Complete and instantaneous mixing occurs at the confluence of the wastewater discharge and the receiving river.
- BOD₅ is treated as conservative, with no biodegradation or decay considered.
- DO is estimated using mass balance only and does not account for re-aeration, nitrification, or downstream oxygen sag processes.
- With exception to the above, other point/diffuse sources along and leading into the receiving river is not explicitly considered.

Given the relatively small channel dimensions of the study river and the screening level nature of the assessment, this assumption is considered appropriate for representing near field mixing behaviour.

4. Results and Discussion

4.1 Wet Market Discharge Profiling

4.1.1 Bangi Wet Market

The discharge quality at Bangi Wet Market showed moderate temporal variability over the sampling period (7:30 a.m. to 4:00 p.m.), reflecting routine produce handling and cleaning activities. The pH remained relatively stable (6.60–7.08), while wastewater temperature increased from 26.0°C to 30.4°C in line with ambient conditions.

DO concentrations were consistently low (1.97–3.21 mg/L), indicating the presence of biodegradable organic matter. BOD₅ and COD ranged from 10–84 mg/L and 24–142 mg/L,

respectively, with sharp peaks at 3:00 p.m. (207 mg/L and 354 mg/L), corresponding to poultry slaughtering and seafood cleaning activities.

SS was generally below 40 mg/L but increased to 96 mg/L at 3.00 p.m, likely due to floor washing and resuspension of settled solids. O&G remained low (<1–3 mg/L) except for a peak of 13 mg/L, while NH₃-N ranged from 0.6 to 10.5 mg/L, reflecting intermittent nitrogen inputs from animal product residues and protein degradation. Faecal coliform counts remained high throughout (9.8×10^3 to 2.05×10^6 counts/100 mL), indicating persistent microbial contamination posing possible risks of waterborne illness to downstream users upon exposure.

Comparison with the Environmental Quality (Industrial Effluent) Regulations 2009 (IER) shows that pH complied with both Standard A and Standard B. BOD₅ generally exceeded Standard A and the peak exceeded both standards. SS was mostly compliant, except the peak value, which exceeded Standard A but remained within Standard B. O&G generally complied with Standard B but exceeded Standard A, while NH₃-N was mostly compliant with Standard B and only marginally exceeded Standard A at its maximum concentration. Overall, Bangi Wet Market discharge is characterised by moderate organic and microbial pollution with short-term spikes associated with cleaning activities.

4.1.2 Mantin Wet Market

Mantin Wet Market exhibited substantially higher pollutant concentrations and stronger temporal variability than Bangi wet market, particularly during early morning poultry slaughtering (5:00 a.m. to 10:00 a.m.). The pH ranged from 4.53 to 6.64, with acidic conditions likely caused by decomposition of blood, fats, and other organic residues. Wastewater temperature remained relatively stable during slaughtering (24.9–26.7°C) before increasing to 31.1°C at 1.30 pm.

DO concentrations were consistently low (2.13–3.17 mg/L), reflecting high oxygen demand. BOD₅ and COD were markedly elevated, ranging from 254–1,775 mg/L and 421–3,093 mg/L, respectively, with peak values at 9:00 a.m. SS concentrations were also high (42–1,000 mg/L), indicating discharge of feathers, tissue residues, and coagulated solids from poultry processing. O&G ranged from 5–96 mg/L, while NH₃-N reached 43.9 mg/L, reflecting fats, protein degradation, and slaughter related waste inputs. Faecal coliform counts were extremely high (5.2×10^5 to 2.34×10^7 counts/100 mL), particularly during peak slaughtering, indicating severe faecal contamination and elevated pathogen risk.

Benchmarking against the IER indicates substantial noncompliance with both Standard A and Standard B for most parameters. The pH failed to comply with Standard A and, at the lower end, Standard B. BOD₅ exceeded both standards throughout, while SS and O&G exceeded permissible limits for most of the sampling period. NH₃-N was only compliant at the lower end during post slaughtering periods but generally exceeded Standard B during intensive operations. Overall, Mantin Wet Market discharge is characterised by severe organic, solids, nutrient, and microbial pollution, with pollutant loading strongly driven by poultry slaughtering activities.

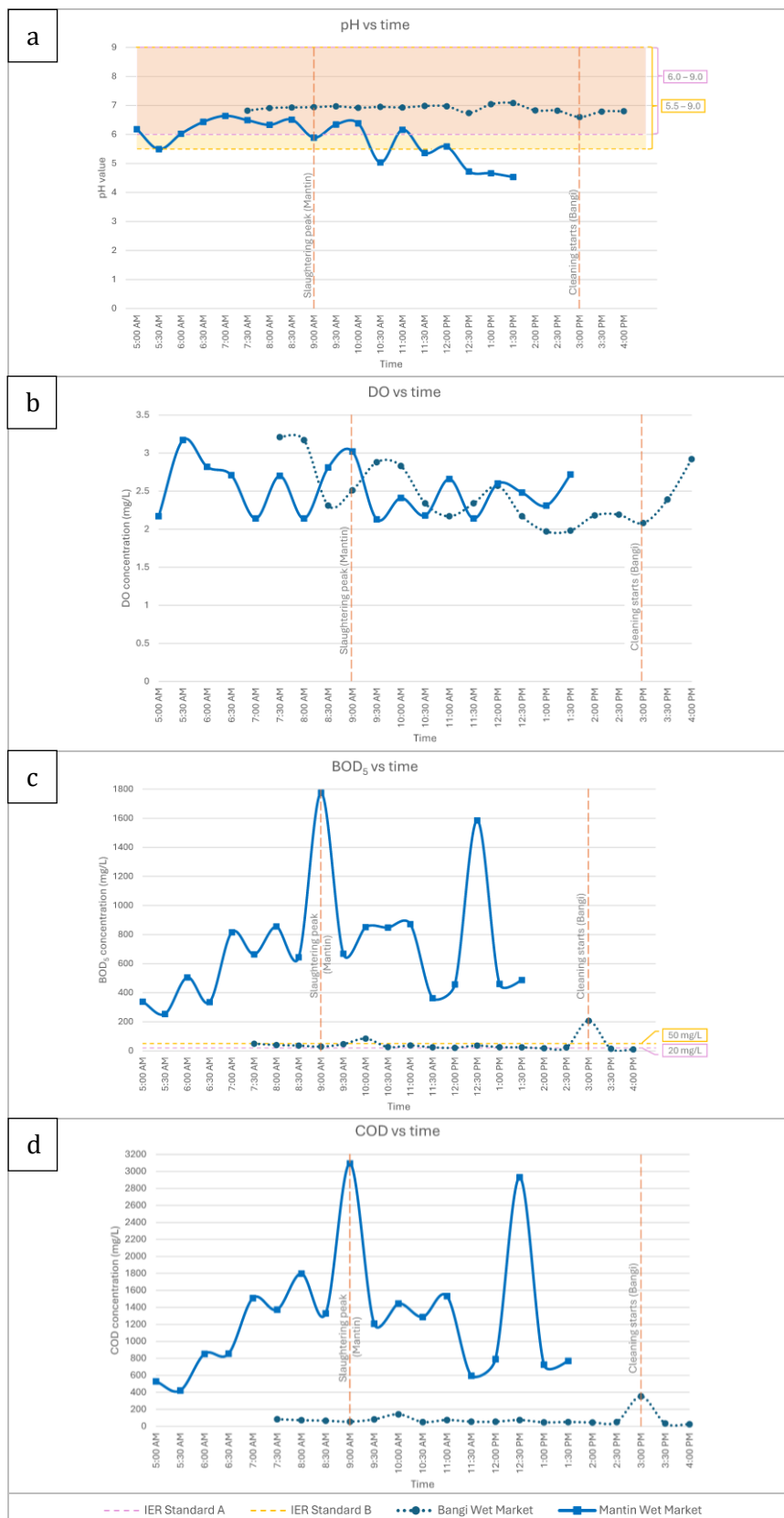


Figure 3. Temporal Trends of Wastewater Discharge Parameters at Mantin and Bangi Wet Markets. (a) pH; (b) dissolved oxygen (DO); (c) biochemical oxygen demand in 5 days at 20 °C (BOD₅); (d) chemical oxygen demand (COD)

Note: IER refers to Environmental Quality (Industrial Effluent) Regulations 2009

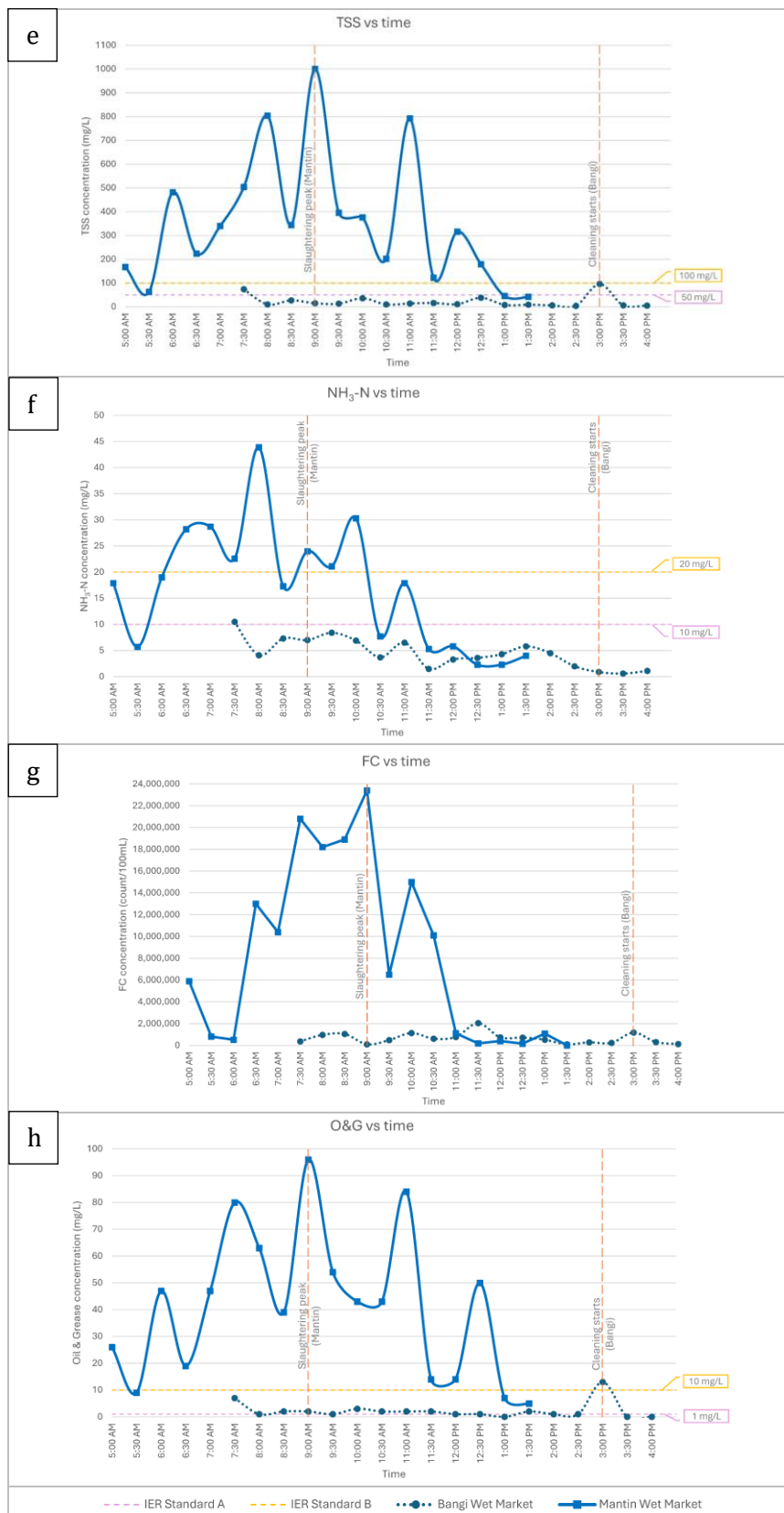


Figure 3 (continued). Temporal Trends of Wastewater Discharge Parameters at Mantin and Bangi Wet Markets. (e) total suspended solids (TSS); (f) ammoniacal nitrogen (NH₃-N); (g) faecal coliform (FC); (h) oil and grease (O&G)

Note: IER refers to Environmental Quality (Industrial Effluent) Regulations 2009

4.2 Mixing Model for Assessing Impacts on Receiving Waters

Table 3 summarises the predicted downstream concentrations under Scenario 1 and Scenario 2. The results were interpreted against the National Water Quality Standards (NWQS) to assess the magnitude of change and the potential for deterioration in river water quality class.

Table 3. Predicted Changes to River Water Quality Relative to Baseline Conditions

Pollutant	Concentration (mg/L)			Remarks
	Baseline	Predicted	Change	
Scenario 1: Peak discharge from wet market during normal flow condition				
DO	5.30	5.30	no significant change	DO level remains within baseline condition (Class IIA/IIB)
BOD	3.00	3.02	0.02	BOD level changes from Class IIA/IIB to Class III
COD	16.00	16.03	0.03	COD level remains within baseline condition (Class IIA/IIB)
TSS	18.00	18.01	0.01	TSS level remains within baseline condition (Class I)
NH ₃ -N	0.40	0.40	no significant change	NH ₃ -N level remains within baseline condition (Class III)
Scenario 2: Peak discharge from wet market during low flow condition				
DO	4.50	4.50	no significant change	DO level remains within baseline condition (Class III)
BOD	3.00	3.76	0.76	BOD level changes from Class IIA/IIB to Class III
COD	16.00	17.31	1.31	COD level remains within baseline condition (Class IIA/IIB)
TSS	47.00	47.41	0.41	TSS level remains within baseline condition (Class IIA/IIB)
NH ₃ -N	0.40	0.41	0.01	NH ₃ -N level remains within baseline condition (Class III)

4.2.1 Discussion

The mixing assessment indicates that peak wet market discharge is likely to have limited impact on most receiving water parameters, with COD, TSS, NH₃-N and DO remaining within their baseline NWQS classes under both normal and low flow conditions. This reflects the relatively small discharge volume compared with river flow and the resulting dilution effect.

BOD was the only parameter predicted to undergo a class shift, changing from Class IIA/IIB to Class III under both scenarios, with a greater increase under low-flow conditions. This is due to the high organic strength of the wet market discharge and the baseline river BOD concentration being close to the class threshold, such that even a small increase after mixing was sufficient to cause class deterioration.

Although the predicted changes for most parameters were minor, the assessment considers only the isolated contribution of wet market discharge and excludes cumulative effects from other pollution sources in the catchment. Overall, the findings suggest that wet market discharges are episodic but potentially important sources of organic loading, particularly under low-flow conditions or in rivers with limited assimilative capacity.

4.3 Proposed Mitigation Measures

The temporal variability of wet market wastewater discharges, coupled with poor segregation and management of biowastes, highlights the need for targeted mitigation measures. Mitigation measures should therefore focus on improved waste segregation, installation of treatment technologies, and stricter regulatory enforcement.

A key control measure is the installation of grease traps or oil interceptors to remove fats, oils and grease before wastewater enters the drainage or sewerage system. Regular maintenance of these systems is necessary at least once a year to ensure effective operation and prevent grease accumulation in drains. Grease traps should be inspected regularly and cleaned before accumulated fats, oils, and grease exceed approximately 25% of the trap volume, as excessive accumulation reduces separation efficiency and increases the risk of blockages. For wet market stalls handling poultry, fish, and meat products, weekly cleaning is recommended, while larger communal grease interceptors may require maintenance every one to two weeks depending on loading conditions [17].

From a regulatory perspective, the discharge of foul or offensive liquids into public drains is prohibited under the Street, Drainage and Building Act 1974, and local authorities may undertake inspections and enforcement against non-compliance. However, given the high strength and variable nature of wet market wastewater, dedicated discharge guidelines or sector specific standards for wet market discharge may be needed to strengthen regulation and control.

Proper drainage infrastructure is also essential to prevent uncontrolled discharge of wastewater into public drains and nearby water bodies. Wastewater generated from fish cleaning, meat processing and stall washing should be collected through floor traps, waste sumps and gully traps, and conveyed via dedicated drainage systems to appropriate treatment facilities or municipal sewer networks where available. Upgrading existing market drainage systems is therefore an important improvement measure.

In addition, awareness programmes for market vendors should be implemented to promote proper wastewater handling, waste segregation and disposal practices. Training and environmental education can help reduce pollution at source and improve sanitation within market areas.

Biowaste treatment and recycling technologies, including biological, physico-chemical and thermochemical processes, also offer potential for reducing pollutant loads and recovering value from wet market wastes. These technologies can support resource recovery, such as fertiliser and bioenergy production, while reducing environmental impacts and reliance on landfill disposal.

5. Conclusion

This study characterised wastewater discharges from two urban wet markets with contrasting operational activities using high resolution (30 min) profiling and assessed potential receiving water impacts using mixing model under normal and lowflow conditions. The findings demonstrate that wet market wastewater quality is strongly influenced by market operations,

with marked temporal variability and pollutant peaks associated with cleaning and slaughtering activities. Bangi Wet Market generally exhibited lower pollutant concentrations with intermittent spikes during washing periods, whereas Mantin Wet Market, where poultry slaughtering was undertaken, produced consistently higher strength wastewater characterised by elevated organic matter, suspended solids, oil and grease, ammoniacal nitrogen, and faecal coliform. These results indicate that slaughtering activities are a key driver of pollutant loading and microbial contamination in wet market effluent.

Comparison with the Environmental Quality (Industrial Effluent) Regulations 2009 showed frequent exceedances at Mantin and occasional exceedances at Bangi, highlighting the inadequacy of untreated discharge. Despite this, the mixing assessment suggested that impacts on most receiving water parameters may remain limited due to river dilution, although BOD was identified as the most sensitive parameter, with potential to cause water quality class deterioration under low flow conditions. Overall, the study highlights wet markets as episodic yet important sources of organic- and pathogen rich wastewater and supports the need for improved wastewater management, treatment, and regulatory control.

References

- [1] Zulkifli A R, Roshadah H and Tunku Khalkausar T F 2012 Control of water pollution from non-industrial premises *Proc. 2nd Int. Conf. Water Resour. (ICWR 2012)* (Langkawi, Malaysia)
- [2] Al-Gheethi A, Ma N L, Rupani P F, Sultana N, Yaakob M A, Radin Mohamed R M S and Soon C F 2021 Biowastes of slaughterhouses and wet markets: An overview of waste management for disease prevention *Environ. Sci. Pollut. Res.*
- [3] Ho K and Chu L M 2019 Characterization of food waste from different sources in Hong Kong *J. Air Waste Manag. Assoc.* **69** 277–288
- [4] Ratnawati R and Trihadiningrum Y 2014 Slaughterhouse solid waste management in Indonesia *J. Biol. Res.* **19** 69–73
- [5] Lee C H, Shari Z, Zairul M, Lawrence D S 2024 Exploring the design framework for food waste recycling awareness in Chow Kit Wet Market
- [6] Nadimpalli M L and Pickering A J 2020 A call for global monitoring of water, sanitation and hygiene in wet markets *The Lancet Planet. Health* **4**(10) e439–e440
- [7] Betoret N, Betoret E and Glicerina V T 2024 Valorization and utilization of food wastes and by-products: Recent trends, innovative technologies and sustainability challenges *Foods* **13**(1) 9
- [8] Salminen E and Rintala J 2002 Anaerobic digestion of organic solid poultry slaughterhouse waste: A review *Bioresour. Technol.* **83** 13–26
- [9] Asses N, Farhat W, Hamdi M and Bouallagui H 2019 Large-scale composting of poultry slaughterhouse processing waste: microbial removal and agricultural biofertilizer application *Process Saf. Environ. Prot.* **124** 128–136
- [10] Purkayastha D and Sarkar S 2022 Sustainable waste management using black soldier fly larvae: A review *Int. J. Environ. Sci. Technol.* **19** 12701–12726
- [11] Mertenat D, Diener S and Zurbrugg C 2019 Black soldier fly biowaste treatment: Assessment of global warming potential *Waste Manag.*
- [12] Yoon Y M, Kim S H, Oh S Y and Kim C H 2014 Potential of anaerobic digestion for material recovery and energy production in waste biomass from a poultry slaughterhouse *Waste Manag.* **34** 204–209
- [13] Sari O F, Ozdemir S, Celebi A 2016 Utilization and management of poultry slaughterhouse waste with new methods *Proc. Eurasia 2016 Waste Management Symposium*
- [14] Shende A D and Pophali G R 2020 Anaerobic treatment of slaughterhouse wastewater: A review *Environ. Sci. Pollut. Res.* **27** 42776–42797
- [15] Selormey G K, Barnes B, Kemausuor F and Darkwah L 2021 Anaerobic digestion of slaughterhouse waste: Effect of selected operational and environmental parameters on anaerobic biodegradability *Rev. Environ. Sci. Biotechnol.* **20** 893–927
- [16] Radin Mohamed R M S, Al-Gheethi A A S, Mohd Apandi N, Mohd Kassim A H and Jais N M 2018 Biomass quality of *Scenedesmus* sp. cultivated in wet market wastewater *Malays. J. Microbiol.* **14** 120–123

- [17] Mahmood B and Bogati R 2021 Assessment of grease traps used in the small-scale food industry: A pilot study *Unitec Research Symposium Proceedings*
- [18] Adetunji A I and Olaniran A O 2021 Treatment of industrial oily wastewater by advanced technologies: A review *Appl. Water Sci.* **11** 98
- [19] Akhtar N A, Kobya M and Gengec E 2025 Chemical and electrochemical coagulation for residual phosphate removal: Efficiency and costs *Environ. Sci.: Water Res. Technol.*
- [20] Koltsova E, Smotraiev R, Nehrii A, Zhekeev M and Ratnaweera H 2025 Mechanisms for removing phosphorus species through sequential coagulation using inorganic coagulants and organic polymers *Water Sci. Technol.* **91(2)** 202–218
- [21] Skrzypczak D, Trzaska K, Mironiuk M, Mikula K, Izydorczyk G, Polomska X, Wiśniewski J, Mielko K, Moustakas K, Chojnacka K 2024 Recent innovations in fertilization with treated digestate from food waste to recover nutrients for arid agricultural fields *Environ. Sci. Pollut. Res. Int.* **31** 41563–41585
- [22] International Water Association and The Nature Conservancy 2021 Nature-Based Solutions for Wastewater Treatment: Factsheets and Case Studies (*London: IWA Publishing*)
- [23] Gusiatin Z M, Antifanow A, Bulat K and Zielińska A 2020 Gasification by-products and their environmental applications (including heavy metal removal) *Environ. Sci. Pollut. Res.*
- [24] International Union for Conservation of Nature 2022 Waste Optimisation: Inspirational Guide for Source Segregated Waste Streams (*Gland: IUCN*)